

Foundations of Retrieval-Augmented Generation

How AI assistants answer from enterprise data: embeddings, vector search and grounded responses

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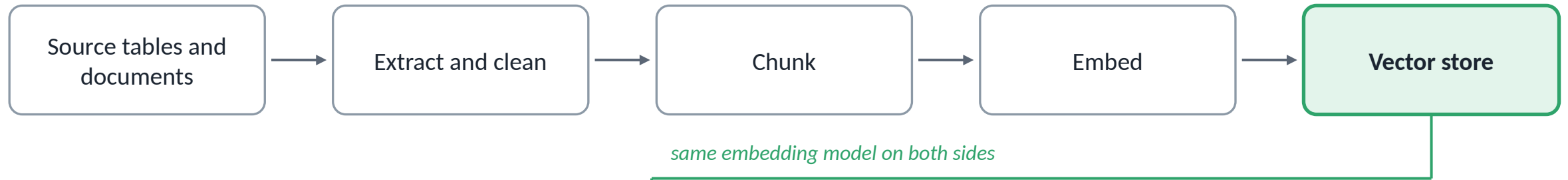
SnowPro Certified · Data Engineering and AI Trainer

Scenario: Neo Nexus, a loyalty assistant for Caffè Nero. Illustrative only: all data, documents and figures are synthetic.

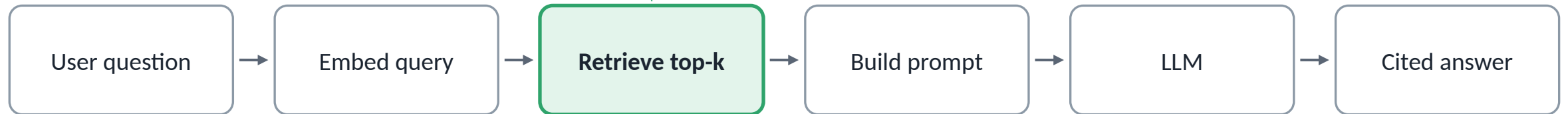


RAG is two pipelines that share one index

OFFLINE • INDEXING (BATCH OR CDC)



ONLINE • PER QUESTION (LATENCY-SENSITIVE)



Data engineering owns the top lane. Most production RAG failures start there.

Similarity search cannot count



Semantic: RAG over text

"Can members redeem a free drink on a delivery order?"

Sources: programme terms, rewards FAQs, store guides

Mechanism: embed, retrieve top-k, ground the answer



router



Analytical: text-to-SQL

"How many free-drink rewards were redeemed in London in Q2?"

Sources: members, rewards, transactions tables

Mechanism: generate SQL over a modelled schema, execute, explain

Module 2 preview: Cortex Search is the managed left route, Cortex Analyst the managed right route, both fed by dbt models.

An LLM reads exactly what you give it

RAW.REWARDS_EXPORT

RWD_ID	RWD_STS	AMT	ISS_DT
RW-0091	R	3.40	20260411
RW-0091	R	3.40	20260411
RW-0107	I		11/04/2026

dbt

MARTS.FCT_REWARDS

reward_id	reward_status	reward_value	issued_date
RW-0091	Redeemed	3.40	2026-04-11
RW-0107	Issued	null	2026-04-11

Duplicates

double loads inflate every count

Cryptic codes

the model must guess that R means Redeemed

Mixed formats

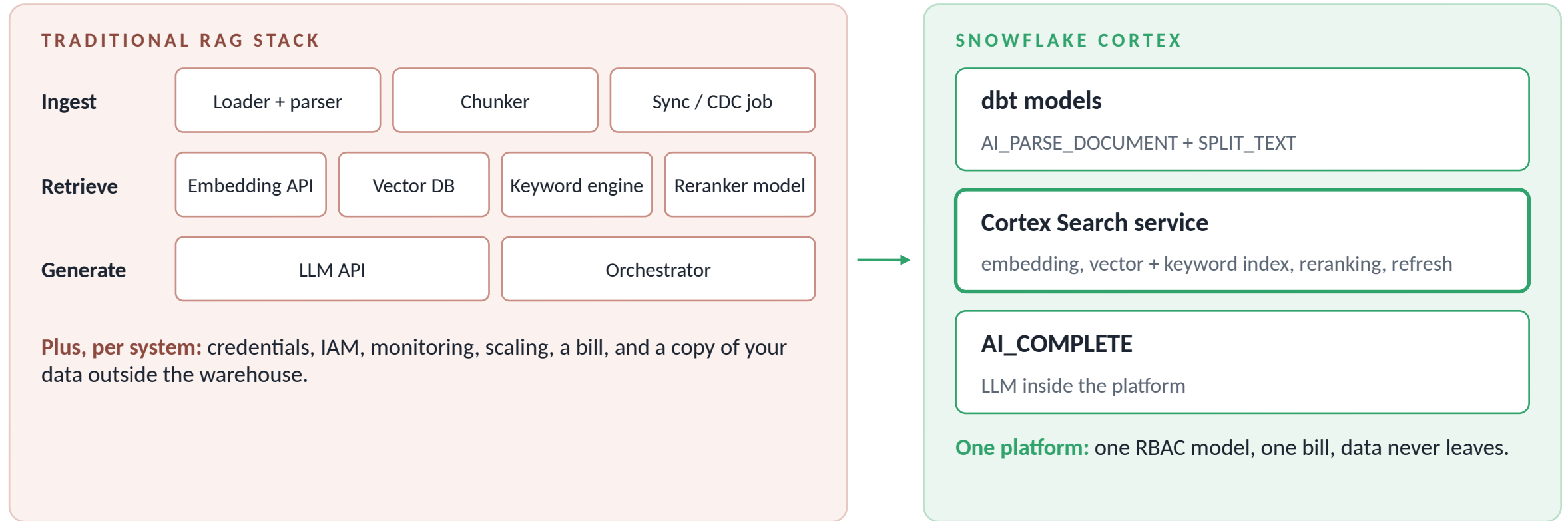
dates and amounts parse inconsistently

No declared grain

one row per what? The LLM cannot know

Principle: the semantic layer an AI reads is a product. Model it, test it and document it like one.

Nine moving parts collapse into three objects



9 → 3

moving parts to build and run

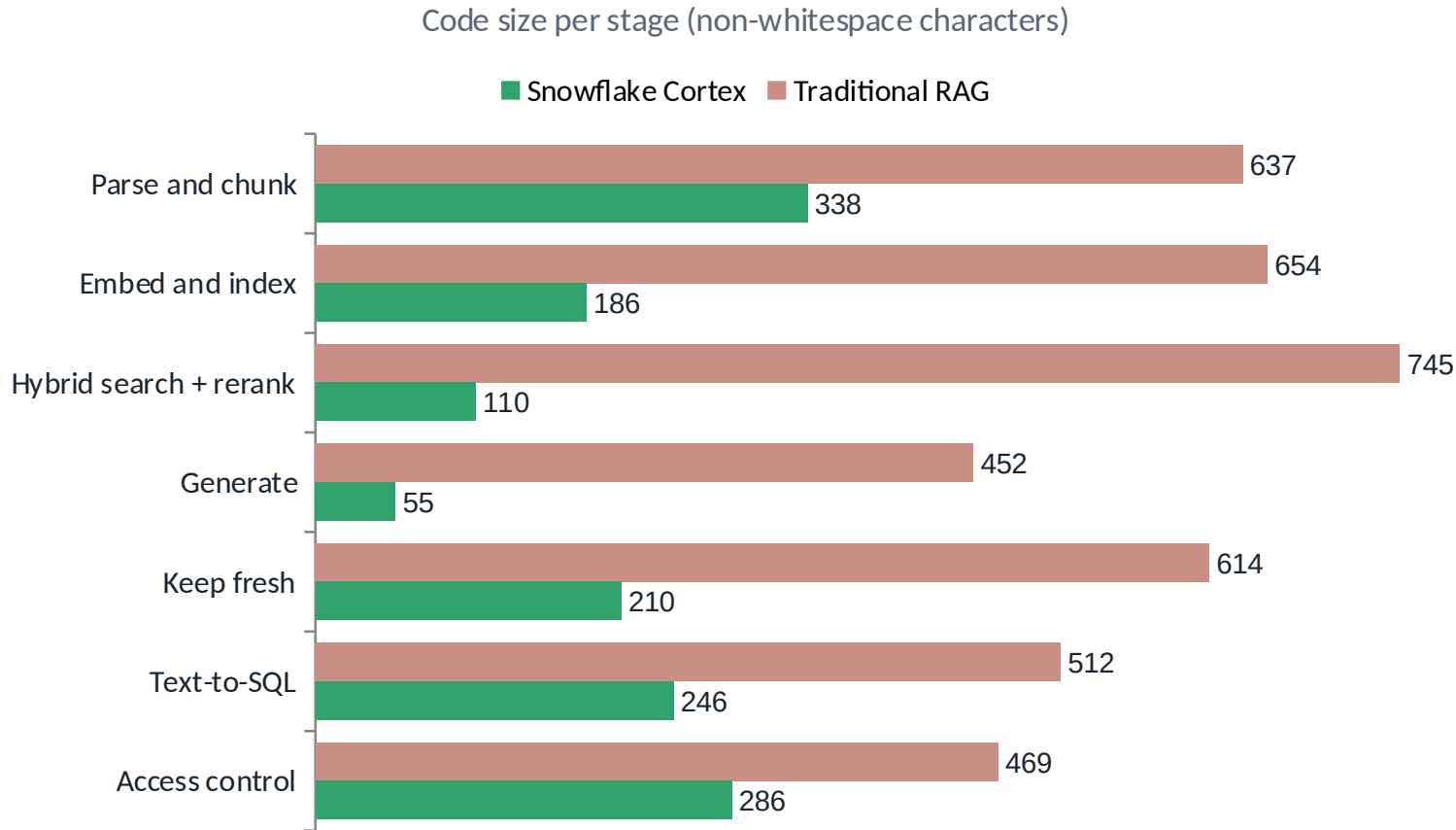
Many → 1

vendors, bills and security boundaries

Scripted → declared

freshness via TARGET_LAG

About two-thirds less code across the same seven stages



4,083 → 1,431

characters of code, about 65% less. In
lines: 115 → 67.

Not counted, traditional side: Elasticsearch,
Airflow, reranker hosting, secrets. **Not**
counted, Cortex side: the semantic view DDL.

One function, many engines: change a string, not a stack

Family	Example AI_COMPLETE model strings
Anthropic Claude	claude-opus-5-5, claude-sonnet-4-5
OpenAI GPT	openai-gpt-5.5, openai-gpt-5-mini
Meta Llama	llama3.3-70b, llama3.1-8b
Mistral	mistral-large3
Google Gemini	gemini-3.1-pro

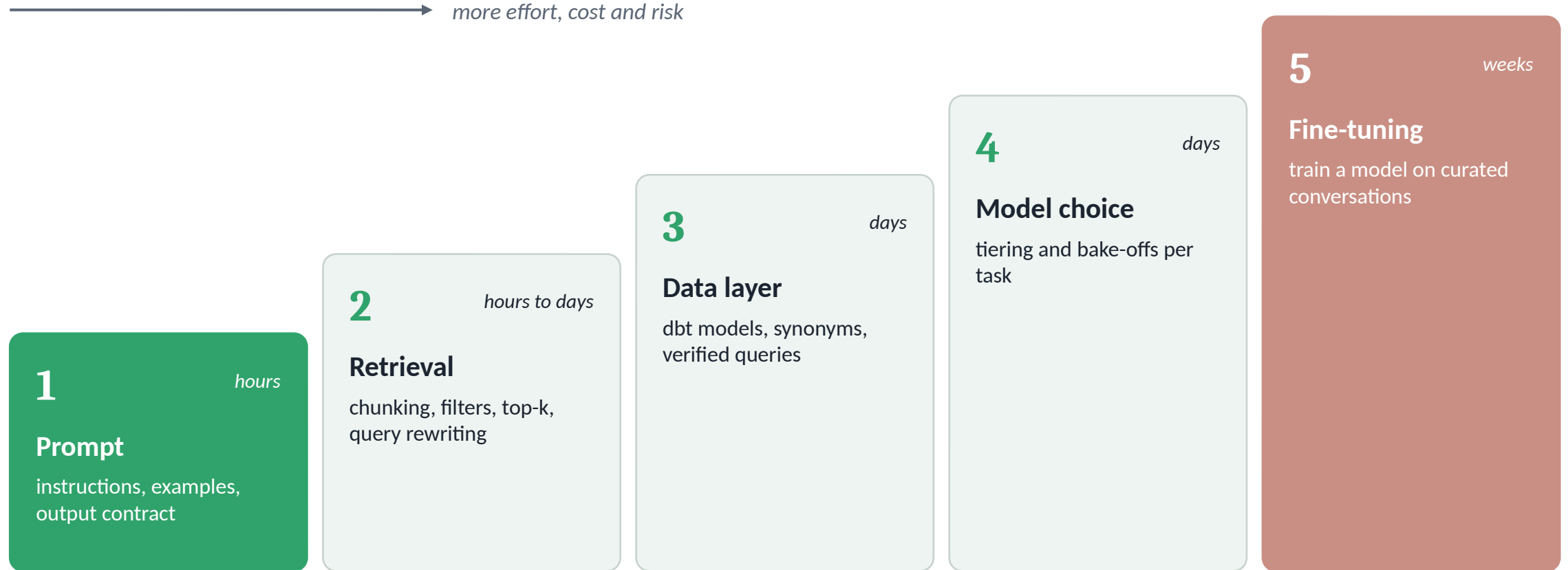
Regional availability varies; some models need cross-region inference.

Module 1: new engine = new SDK, key and infrastructure. **Cortex:** new engine = new string.

CORTEX · SQL

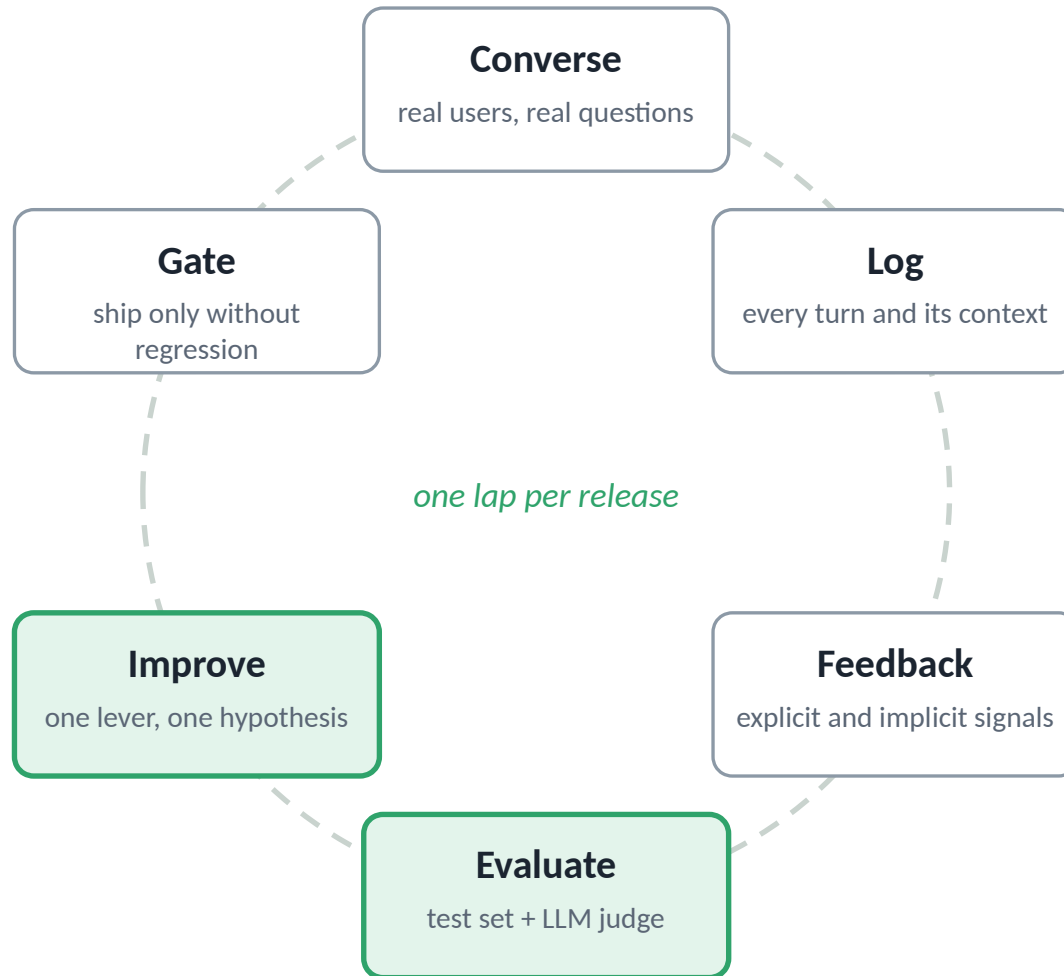
```
-- same prompts, three engines,  
-- one query: a model bake-off  
select  
    question_id,  
    ai_complete('claude-sonnet-4-5',  
                prompt) as claude,  
    ai_complete('openai-gpt-5-mini',  
                prompt) as gpt,  
    ai_complete('llama3.3-70b',  
                prompt) as llama  
from eval.golden_prompts;
```

Climb from the cheapest lever to the most expensive



Start at rung 1. Climb only when the evidence says the lower rung cannot fix it.

Conversations become evidence; evidence becomes changes



Part 1 • Capture

Log and Feedback: what happened in each turn, and was it good?

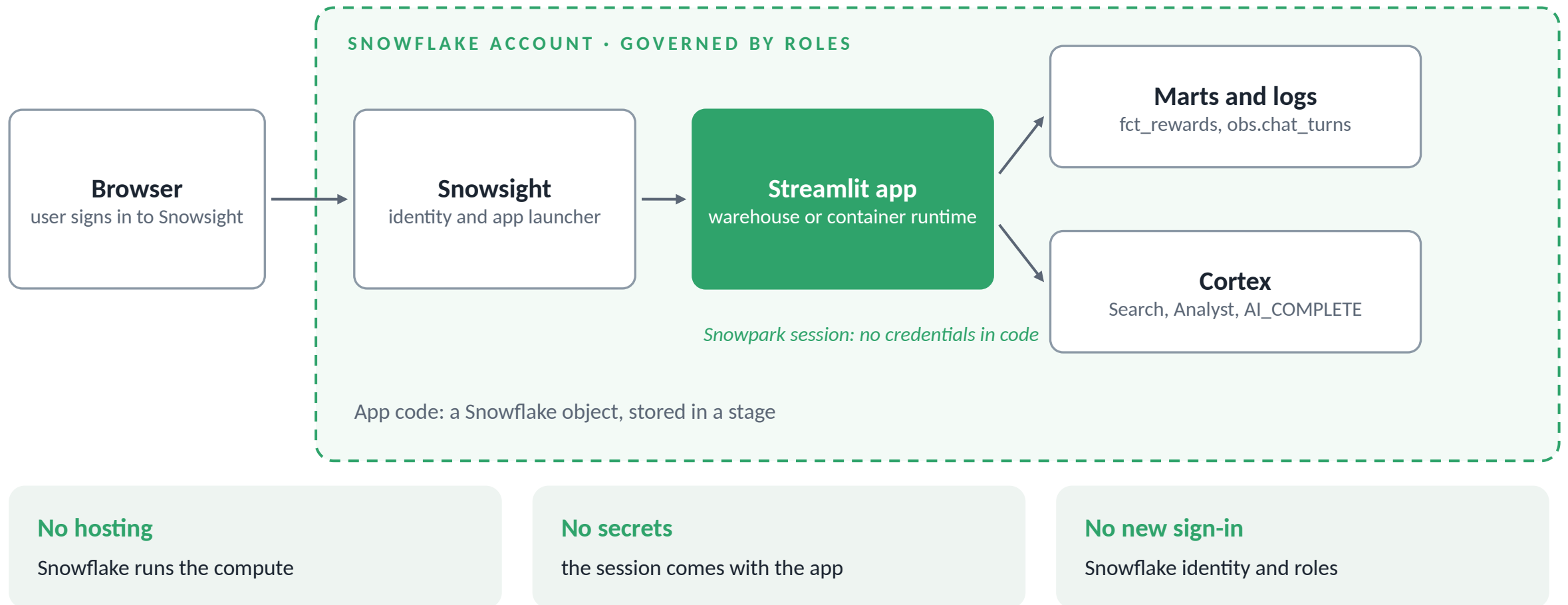
Part 2 • Evaluate

Evaluate: a repeatable score for the current version, and a map of its failures.

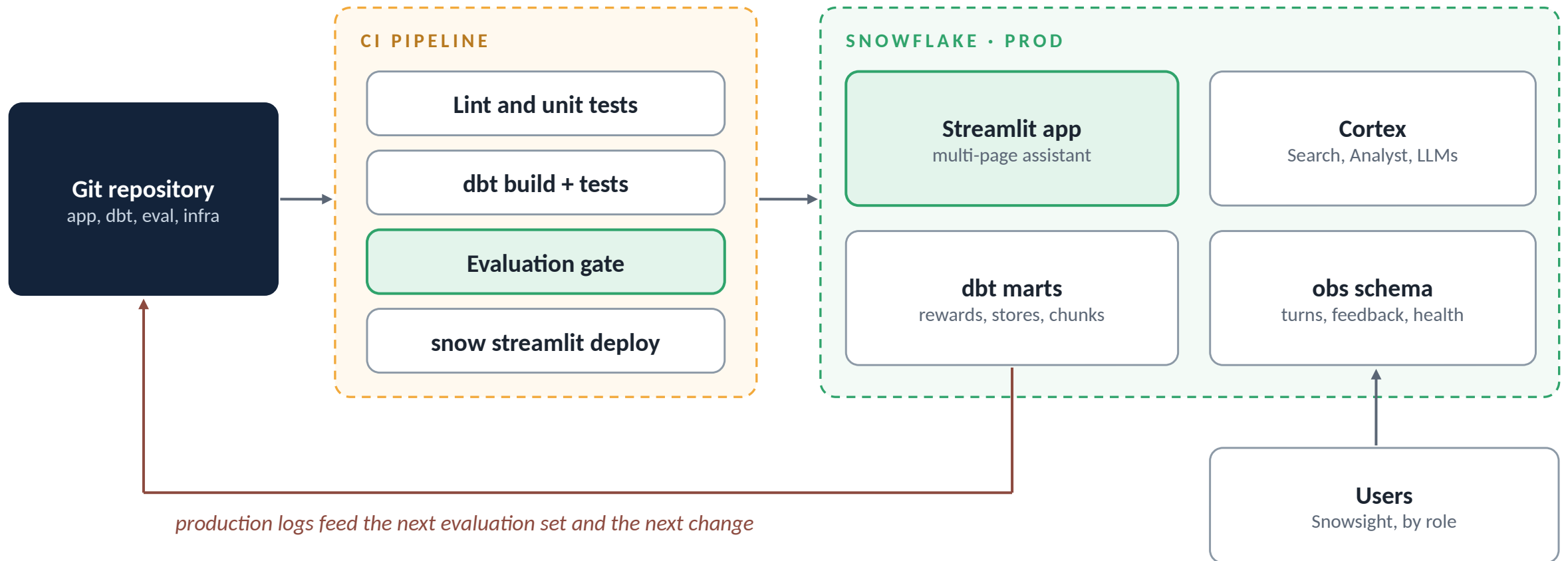
Part 3 • Optimise

Improve and Gate: pull the cheapest effective lever and prove it helped.

The app runs inside the platform that holds the data

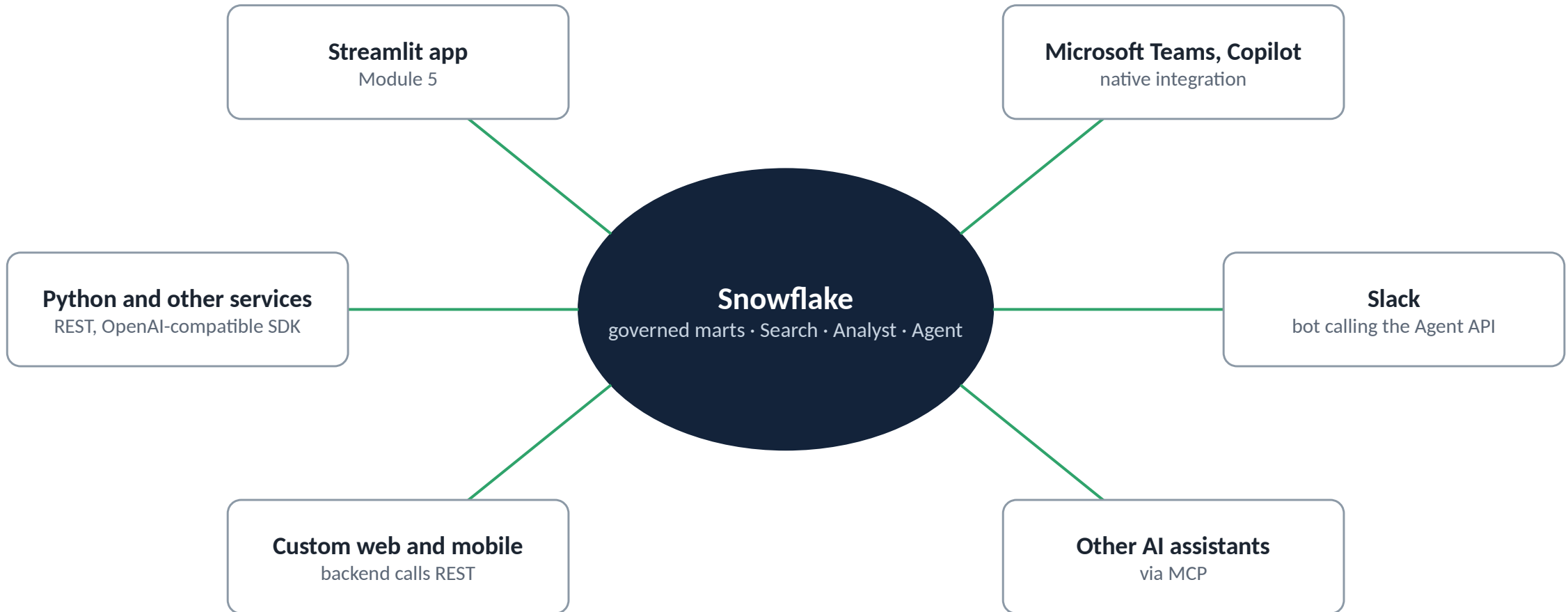


Git to pipeline to Snowflake, with a loop back from production



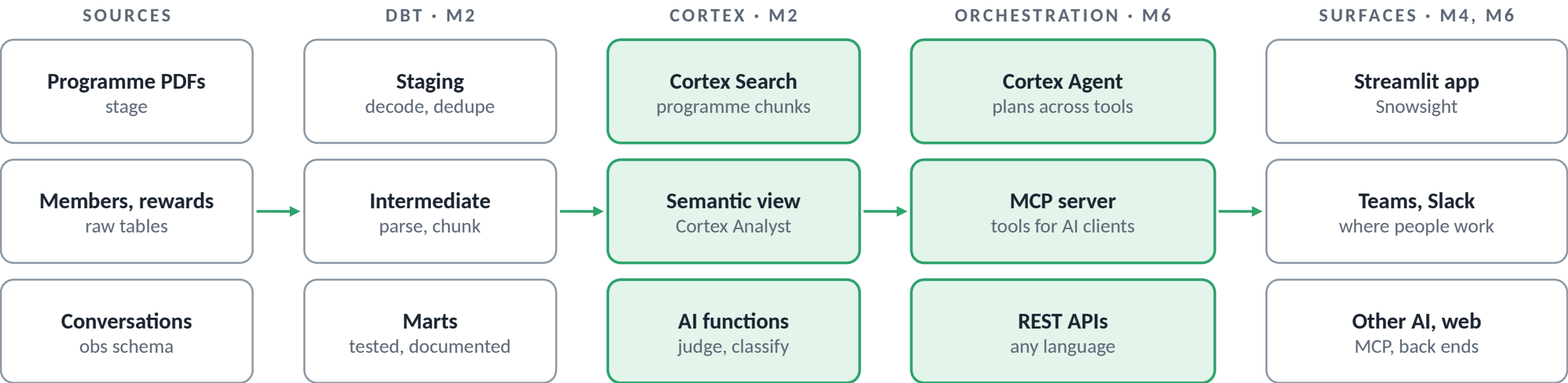
Every box here is covered today.

Move the question to the data, not the data to the question



One governed hub; every surface sends questions in and gets answers out.

Reference architecture for a governed enterprise assistant



Learn • M3 log every turn • feedback • evaluation set • LLM judge • failure classification • optimisation ladder

Ship • M5 Git • CI • evaluation gate • DEV, TEST, PROD • least-privilege roles • health view, alerts, runbook

From a vector database to a governed AI platform



The lasting lesson: models change every few months. Governed data, honest evaluation and disciplined delivery are what make an assistant worth trusting.